



Eco-driving key factors that influence fuel consumption in heavy-truck fleets: A Colombian case

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ABSTRACT

This research identifies key variables that influence fuel consumption that might be improved through eco-driving training programs under three circumstances that have been scarcely studied before: (a) heavy- and medium-duty truck fleets, (b) long-distance freight transport, and (c) the Latin American region. Based on statistical analyses that include multivariate regression of operational variables on fuel consumption, the impacts of an eco-driving training campaign were measured by comparing ex ante and ex post data. Operational variables are grouped into driving errors, trip conditions, driver behavior, driver profile, and vehicle attributes.

The methodology is applied in a freight fleet with nationwide transport operations located in Colombia, where the steepness of its roads plays an important role in fuel consumption. The fleet, composed of 18 trucks, is equipped with state-of-the-art real-time data logger systems. During four months, 517 trips traveling a total distance of 292,512 km and carrying a total of 10,034 tons were analyzed.

The results show a baseline average fuel consumption (FC) of 1.716 liters per ton-100 km. A different logistics performance indicator, which measures FC in liters per ton transported each 100 km, shows an average of 3.115. After the eco-driving campaign, reductions of 6.8% and 5.5% were obtained. Drivers' experience, driving errors, average speed, and weight-capacity ratio, among others, were found to be highly relevant to FC. In particular, driving errors such as acceleration, braking and speed excesses are the most sensitive to eco-driving training, showing reductions of up to 96% on the average number of events per trip.

1. Introduction

In 2014, oil represented 31% of the total global primary energy supply, representing 34% of the global CO₂ emissions (IEA, 2016). At the same time, transportation represented 23% of these emissions, with a noteworthy 71% increase since 1990. This pollution growth was mainly driven by the road sector, which represented three quarters of the transport emissions, including the heavy-truck industry, which moves a huge portion of the freight worldwide. In 2014, Latin America represented 4% of global CO₂ emissions and

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Table 1

Relevant issues covered by the recent eco-driving literature.

Author	Variables							Scope			Analysis and results features									
	VP	VE	Em	RC	LE	DC	DP	DS	Urb	LD	LS	SM	ER	MR	Frg	RT	Place	*Others		
Current research	*			*	*	*	*			*		*	*	*	*	*	Colombia			
Ayyildiz et al. (2017)	*	*	*		*				*			*	*	*	*	*	China			
Xu et al. (2017)	*	*	*	*		*			*				*	*		*	U.S.	b		
Ma et al. (2015)	*	*			*	*			*				*			*	China			
De Abreu e Silva et al. (2015)	*			*		*	*		*		*	*	*	*		*	Portugal	b, d, e		
Walnum and Simonsen (2015)	*			*	*					*		*	*	*	*	*	Norway	d, e		
Lai (2015)							*		*			*	*	*		*	Taiwan	s		
Sullman et al. (2015)	*						*	*								*	UK-Finland	b, s		
Rolim et al. (2014)							*		*		*		*	*		*	Portugal	b		
Caulfield et al. (2014)			*	*	*		*		*		*	*	*	*		*	The Netherlands			
Birrel et al. (2014)	*								*			*	*			*	Several			
Wählberg (2007)	*				*		*		*				*			*	Sweden	b		
Stromberg and Karlsson (2013)							*		*		*					*	Sweden	b, s		
Delhomme et al. (2013)							*		*			*	*			*	France	s		
Beusen et al. (2009)	*								*			*	*			*	Belgium			
Rutty et al. (2013)	*		*						*				*			*	Canada			
Ando and Nishihori (2012)			*				*		*			*	*			*	Japan			
Liimatainen (2011)	*			*					*			*	*	*		*	Finland	b, s		

VP: vehicle parameters.

VE: vehicle or engine models.

Em: emissions.

RC: road/route infrastructure & traffic conditions.

LE: load and weight effects.

DC: driving cycles/route.

DP: driver profile.

DS: chassis dynamometers tests or simulations.

Urb: urban or suburban.

LD: long distances.

LS: feedback logging systems effects.

SM: statistical models/hypothesis tests.

ER: empirical results.

MR: managerial recommendations.

Frg: freight transport.

RT: real time data.

Comments: b: buses, d: daily data, s: surveys, e: elasticity.

presented a 98.1% increase in emissions from fuel consumption (FC) since 1990 (IEA, 2016). Consequently, countries such as Colombia show a huge dependence on fossil fuels at 76% of the total energy consumption (World Bank, 2017), and more than 90% of CO₂ emissions by the transportation sector originates from road transport (IDEAM, 2016), of which 50% is due to freight transport (MinTransporte, 2014). To counteract this finding, eco-driving has emerged as one of the most effective strategies to decrease fuel consumption and emissions, showing savings that range from 0% to 18% (Stromberg and Karlsson, 2013; Rutty et al., 2013; Xu et al., 2017).

Sanguinetti et al. (2017) have identified six classes of eco-driving actions: driving, cabin comfort, trip planning, load management, fueling, and maintenance. This research focuses on the driving class, in which eco-driving training programs for drivers are the key enabler to enhance safer driving and lower fuel consumption. The training is conceptual and practical to ensure knowledge transfer to drivers, but their success depends on how firms incorporate assessment tools and metrics to maintain eco-driving styles for their drivers. Further details about eco-driving techniques are briefly noted by several researchers (see Wählberg, 2007; Birrel et al., 2014).

There is a vast body of literature concerning fuel consumption and energy efficiency estimates in vehicle fleets. Diverse approaches exist to understand and analyze the relevant components of fuel-economy functions that are part of vehicle-engine models (Mensing et al., 2014; Ma et al., 2015; Yu et al., 2016), driving-style-related models (Delhomme et al., 2013), selection and use of driving cycles (Mensing et al., 2014), and eco-driving assessments regarding their impact on emissions and fuel consumption (De Abreu e Silva et al., 2015). On the other hand, data validation methods vary from technically controlled tests on chassis dynamometers, to empirical and statistical analysis of real-world data acquired with specialized technologies. Table 1 shows a summary of key issues considered by recent studies related to eco-driving programs to assess and improve fuel consumption. Studies that focused on logging systems and feedback to users are excluded from this analysis; however, readers are referred to the literature review of Rolim et al. (2014) for further details on this specific topic.

The classification presented in Table 1 includes a first set of operational factors that have an impact on fuel consumption. These factors consist of vehicle parameters (i.e., vehicle attributes important for the fuel consumption such as weight, age, and power), route conditions (e.g., driving cycle, road conditions, and level of service), emissions (e.g., emission-based models), vehicle or engine

models (e.g., vehicle dynamic models), load effects, and driver profile (i.e., features directly related to the driver such as age, education, training, and experience and influence driver behavior). The second set of factors distinguishes data collected from: (i) dynamometer tests or simulation data; (ii) urban, suburban, and rural empirical data; and (iii) long distance empirical data. Finally, the third set identifies methodology features such as technology feedback issues, statistical methods, optimization modeling, empirical results, managerial recommendations, and freight application, as well as the place where the study is conducted. Certain comments are added to highlight applications such as buses, “b”; the type of data collection, daily collected data via technology “d”; through surveys, “s”; and the type of analyses performed, such as elasticity analysis, “e”. The “driving errors” concept is not included in Table 1, since none of the articles reviewed explicitly considered it.

Vehicle-engine models and emissions have been widely studied in the literature (Faris et al., 2011; Zhou et al., 2016); however, the remaining factors have garnered considerably less focus. Recently, Sullman et al. (2015) studied the effects of eco-driving actions with simulation; however, to maintain a controlled environment, they ignored the influence of the operational variables related to logging systems, their use and its effects on user behavior, and consequently, on fuel consumption. Conversely, Rolim et al. (2014) introduced a review focused on generic logging systems and their effects on users, without considering the effects of specific features and the behaviors of drivers. According to Table 1, studies that include real-time data collection have been mainly implemented in Europe, the U.S., China and Japan; furthermore, hilly topographies have only been considered by Walnum and Simonsen (2015). Regarding freight transport applications on eco-driving programs, Walnum and Simonsen (2015) and Ayyildiz et al. (2017) are among the few that measure the effect of eco-driving training programs on freight heavy fleets.

However, there is no similar study performed in South America, where the reduction of pollution and fuel consumption are a major challenge to road freight transportation sector. Other related issues of Latin America are growing urbanization and motorization rates, huge low- and middle-income classes, fragmented logistics operations, challenging topographical issues, poor infrastructure and lack of proper policy making (Dobbs et al., 2016). These aspects hinder the possibility to adopt conclusions from studies performed in other locations. The latter formulates the opportunity to build novel analyses to gain knowledge about how eco-driving can impact fuel consumption through factors linked to infrastructure, vehicles, staff and other controllable attributes but based on data from the reality (i.e., country, time, firm, and segment) under study (Birrel et al., 2014; Ando and Nishihori, 2012; Xu et al., 2017). Consequently, company and country-specific features strongly matter in analyzing how driving issues may impact the overall performance (Walnum and Simonsen, 2015).

Finally, eco-driving training effects decrease with time due to the natural tendency of drivers to return to previous driving behaviors (Beusen et al., 2009). The implementation of incentive programs has shown to have a significant effect on fuel consumption and road safety in the long term (CEPAL, 2010). Furthermore, drivers receive feedback about their performance to improve and to gain internal recognition and additional economic support to meet strategic goals (Liimatainen, 2011). Despite the inherent challenges for the application of these incentive programs (i.e., complexity in deployment, lack of accurate consumption data, and lack of information on how to operate the program), eco-driving training is recommended to firms with huge volumes and highly frequent flows to standardize effective driving behaviors and good practices among their staff (Browne and Coyle, 2004).

Therefore, the following research questions are posed to close the aforementioned gaps: (1) Which variables are significant to determine fuel consumption in a heavy-duty truck fleet? (2) What are the immediate impacts on these variables after performing an eco-driving training program? And (3) How could these results be used to promote long-term efficient driving behaviors in firms with heavy-duty fleets? This study proposes a statistical methodology to identify the relevant variables that influence fuel consumption and to evaluate their impact after an eco-driving training campaign. The proposed methodology is tested in a freight fleet with nationwide operations, which is located in Colombia and equipped with logging systems to acquire real-time data on fuel consumption and other operational variables.

The remainder of this paper is organized as follows; Section 2 describes the methodology followed in this work. Section 3 presents the results, beginning with data characterization and followed by the statistical model analyses and discussion. Finally, Section 4 states conclusions, corporate recommendations and limitations.

2. Methodology

The proposed methodology is summarized in Fig. 1. Briefly, an analysis unit is selected in such a manner that it allows the answering of the research questions. This fleet is a heavy/medium freight with long distance operations in a Latin American region. Then, a baseline period is measured and compared to a post-campaign period having an eco-driving training campaign in between. The pre- and post-periods consist of instrumentation, data collection and data pre-processing. The training campaign concentrates on teaching eco-driving concepts, techniques and principles to distinct levels of the organization. In the data analytics step, descriptive analyses, multivariate regressions and hypothesis testing are performed to compare the two data sets.

Regarding the data collected, unlike passenger transportation, freight transportation operation handles routes that are not scheduled in a cyclic manner and drivers may be assigned to different vehicles and routes according to demand. Furthermore, the cargo and vehicle weights may become relevant and they depend on vehicle type and whether it is a loaded or empty. Thus, deeming these features, this work analyzes a set of operational variables not studied all together before. These variables are: driving errors (e.g., excessive speed, acceleration, and braking), trip conditions (e.g., trip length and road topography), driver behavior (e.g., engine speed in revolutions per minute [RPM] and vehicle speed [km/h]), driver profile (e.g., age, experience, and level of education) and operational needs (e.g., load and vehicle attributes).

Next subsections describe with more detail each step of the methodology.

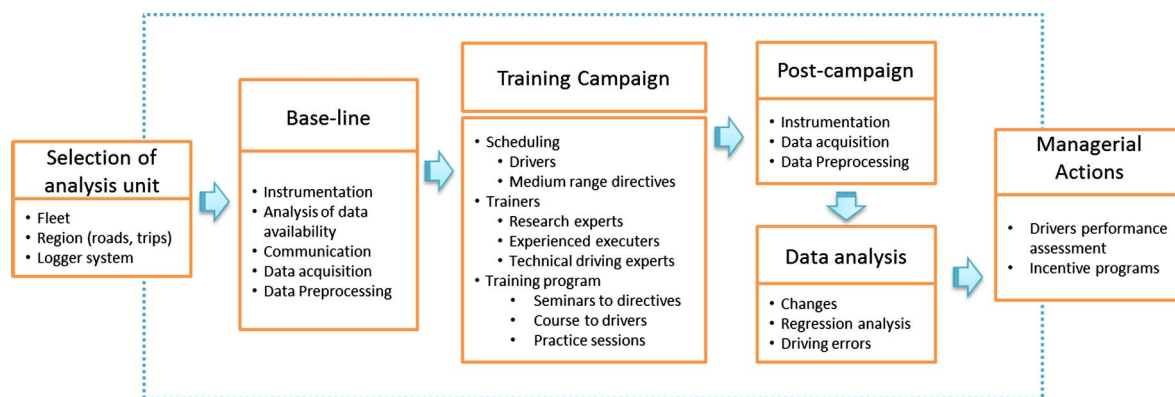


Fig. 1. Eco-driving methodology scheme.

2.1. The region

The Colombian region was chosen as an interesting case study because one-third of its surface is mountainous due to the presence of three Andean ranges. Hilly inhabitable areas have altitudes ranging between 2000 and 3000 meters above sea level (masl) with peaks reaching above 5000 masl, while valleys are approximately 1000 masl. In this varied territory, Colombia's main road corridors run along ridges; however, they must cross at least one ridge at a point, with very curvy sections and challenging road slopes that can achieve a maximum of 13%. Colombia possesses 18,000 km of primary roads (8.5% of the national road network) and 45,000 km of secondary roads (32% of national road network) in poor conditions (see Fig. 2) to move approximately 220 million tons of cargo (MinTransp, 2015). Most of the logistics movements are among the main cities and the ports, located on both the Pacific and Atlantic oceans.



Fig. 2. Main road corridors. Based on DNP (2010).

Table 2
Vehicle specifications and sample composition.

Type	T2	S2	M3	M5_1	M5_2
Brand	Chevrolet	Hino	International	Freightliner	International
Line	FRR Forward 7.8 T	DUTRO 500	Workstar 7600	CL120	Workstar 7600
Engine	Isuzu 4HK1-TCS Turbo	J08E-UD	Cummins ISM 320 V	Cummins ISX 450	Cummins ISM 410
Axles	2	2	3	5	5
Class	C5	C4	C7	C8	C8
Year	2012	2012	2012	2008	2012
Cylinder (cubic centimeters)	5193	4945	8700	14945	11000
Power (Horsepower, HP)	210	247	305	450	400
Gross weight (tons)	10.40	16.93	39.3	54.54	49.1
Empty weight (tons)	5.52	7.73	15.3	17.46	17.1
Capacity (tons)	7	9.2	24	32	34

2.2. The freight fleet

The company's business unit that owns the fleet produces ceramic goods. The company has 38 trucks and employs approximately 1000 independent owner-operator drivers to haul its products. This research distinguishes no type of stock keeping unit (SKU) and only the total load weight since analyses are performed at an aggregate level. Vehicle specifications and the sample composition are summarized in Table 2.

2.3. Eco-driving training campaign

The training campaign shown in Fig. 3 was performed at three key organizational levels: (i) at the operational level, 50 drivers who took the program in two cities at different times. This training program was taught by instructors from a national technical institution (i.e., SENA). The instructors were previously trained by the authors, who also shared the material that was developed; (ii) at the tactical level, middle managers, who received lectures from experts from a couple of Latin America universities on how to develop high-performance eco-driving and follow key principles; and (iii) at the strategic level, strategic top decision makers, who discussed incentive programs and how to create metrics aligned to eco-driving trainings in lower levels with experts. The common means among distinct levels was the material that supported the training.

Fig. 3 displays the content of eco-driving workshops. This material includes an eco-driving manual for drivers and middle managers, benchmark research for middle and top managers, and an analysis of key performance indicators (KPI) to assess drivers' performance, its relationships with fuel consumption and the means to incentivize better performance for top decision managers. The practical session on the road was conducted for the participant drivers.

2.4. Data collection

Data with the greatest level of detail before and after the eco-driving training campaign are necessary to perform an in-depth analysis. *Baseline* data collection was conducted during June and July 2014. Company's drivers were trained in August 2014, and *post-campaign* data collection was performed during September and October 2014.

In the baseline data collection period, 32 vehicles were instrumented and calibrated by a technological third-party partner that continuously monitored the fleet and several driving and vehicle variables. Nevertheless, only 19 diesel vehicles (50% of the fleet) were considered, given their higher frequency of deliveries, the usage of main routes and the quantity of freight transported during

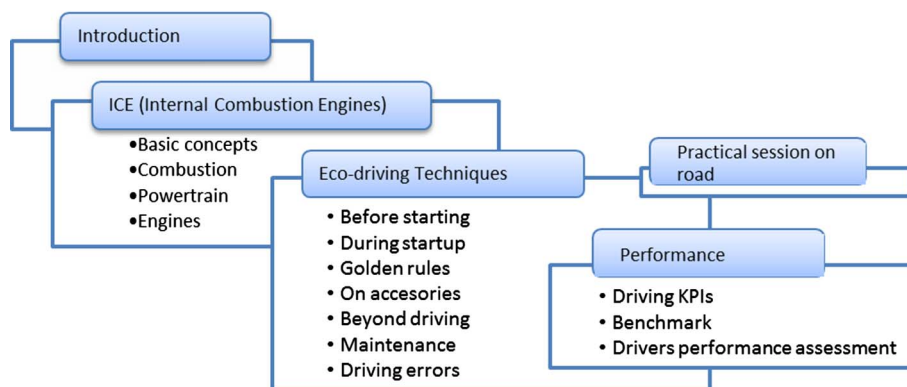


Fig. 3. Content of the eco-driving training material.

the period. These vehicles represent 277 trips and were operated by 18 drivers in total. During the post-training data collection period, the same number of vehicles and drivers (i.e., 19 vehicles and 18 drivers, respectively) were used in the sample, but only 240 trips were performed. The main reason for this reduction is explained by the organizational decisions (e.g., re-allocations) during the data collection period.

To collect the data in both periods, the system used to log vehicle data is a vehicle display output (VDO) Mix Telematics FM 3306 Communicator®, an in-vehicle onboard computer with standard controller area network (CAN) communication protocol and with a main electronic module of the vehicle, a web interface, an internal global positioning system (GPS) receiver and a 3G modem for data transmission. This system is able to gather data from more than 70 vehicle variables simultaneously at a frequency of 1 Hertz. In this study, the collected data include date, hour, distance, run time, speed, time that the engine is on, fuel consumption (FC), journey duration, journey start time, journey end time, driver's name, driver's ID, vehicle's ID and driving errors.

In general, three types of data were collected: (1) driver profile (i.e. age, years of experience, and educational level), (2) vehicle specifications (e.g. brand, weight, age and capacity) and (3) performance on the road (e.g. fuel consumption and driving errors). The data were obtained from two sources: (i) corporate freight management system (CFMS), which represents 20 variables related to driver identification; driver profile; vehicle specifications; direct truck scale readings inside facilities (to monitor vehicle utilization); and trip features; and (ii) Third-party fleet control system (TPFCS), which represents data such as (a) reports per trip consisting of time, distance and speed variables, (b) fuel consumption report, and (c) report of driving errors, composed of number of occurrences, maximum value and duration of the following events: excessive speed above 80 km/h, excessive RPM (vehicle dependent), abrupt acceleration above 10 km/h s, abrupt braking above 14 km/h s, excessive idle time above 5 min with engine speed below 1200 RPM, and time outside the green band (OGB), which occurs when a combination of low RPM and high speed occurs. Table 3 provides the specifications of driving errors from the TPFCS system per vehicle brand.

2.5. Discussion on data processing

In this study, the basic unit handled is a “trip”. There are three types of trips: (a) a duty travel or freight movement, (b) an (empty) vehicle movement between plants, and (c) a plant inbound movement of the vehicle. However, the TPFCS creates a vehicle trip from the moment the vehicle is turned on, until it is turned off. Although the technological system possesses data records from every second, a preprocessing stage was performed to match TPFCS's trips under the trip definition handled.

Additionally, a double check process was performed with internal transportation orders to prove the existence of the orders. This process reduced the complete sample of records to only truly loaded movements. Finally, extreme outlier trips identified in the regression analysis were later excluded from the sample as discussed below. Consequently, only data from reliable trips with identifiable start and end points or times, driven by trained drivers present at both moments (before and after the training campaign) and in trucks properly calibrated and completely instrumented vehicles were considered in this study.

2.6. Statistical analysis

Although fuel consumption (FC) estimation is not the ultimate metric of eco-driving campaigns or of this study, it helps to identify the most relevant variables that affect it. Furthermore, it aims to provide recommendations to fleet management decision makers. Fuel consumption was measured in liters per ton-100 km or [l/t-100 km] and in liters per transported ton each 100 km or [l/100 km-transported tons] or [l/100 km-tt]. The first metric is mostly used from the vehicle perspective, while the second metric is applied on the logistics perspective. Therefore, this research applies statistical analyses to describe the data sets, hypothesis testing and a series of regression models to explain the variation on FC.

In fact, various regression models concerning the multiple variables described in Table 6 and their combinations/transformations were considered. For example, the log of fuel consumption, age and distance, as well as the inverse of distance and the percentage of time outside of the green band (% OGB), were considered. In addition, the interaction of three measurements of the same driving error, such as the number of events during which a driver exceeded the speed limit, the duration and the worst case, were analyzed. Moreover, after performing multicollinearity analyses, many of these potential combinations were removed or reduced (e.g., total weight/vehicle capacity, and TW/Cap, was replaced by only one of the elements but not both). Due to the natural correlation of the number of events, the duration, and the worst case of each driving error, the regression models usually contain, at most, one of them.

Table 3
Driving errors definition.

Brand	Type	Excessive RPM	Outside green band (OGB)
Freightliner	Heavy-duty (M5_1)	> 1.900 RPM	< 1.200 RPM or > 1.700 RPM and > 25 km/h
International	Heavy-duty (M3, M5_2)	> 1.900 RPM	< 1.200 RPM or > 1.700 RPM and > 25 km/h
Chevrolet	Medium-duty Turbo (T2)	> 3.100 RPM	< 1.500 RPM or > 2.600 RPM and > 25 km/h
Hino	Medium-duty (S2)	> 3.100 RPM	< 1.100 RPM or > 2.200 RPM and > 25 km/h

Table 4
Descriptive statistics per fleet.

Type	T2	S2	M3	M5_1	M5_2
<i>Baseline descriptive statistics</i>					
Sample size	4	1	4	1	9
Trips (%)	9.38	36.10	20.58	5.42	28.52
# of tons transported	6.85	8.86	19.63	31.41	30.56
Vehicle utilization (%)	97.80	97.50	81.79	98.16	90.59
<i>Post campaign descriptive statistics</i>					
Sample size	2	5	2	3	7
Trips (%)	13.75	25.42	5.00	16.67	39.17
# of tons transported	6.37	8.37	17.33	29.15	30.28
Vehicle utilization (%)	90.96	91.02	72.20	91.09	89.05

3. Results and discussion

During the collection periods, trips represented a total of 292,512 km traveled in more than 7000 driving hours. Descriptive statistics of the baseline and the post-training period are provided in Table 4. If analyses per route are performed, sample sizes will range from 5 to 32 trips with covering distances ranging from 3000 km to 14,700 km. It is important to highlight that uncontrolled factors (e.g., weather, traffic conditions, drivers' health, pavement conditions, and safety conditions) may have influenced results. However, this uncertainty will always be present, and it may be expected and included as normal errors in advance with the support of statistical modeling. Therefore, statistical analyses were chosen to understand and characterize the significant factors that describe FC of the focal firm.

Furthermore, certain organizational changes between the two collection periods occurred. These variations consisted of the following: reduction of fleet size, staff rotation and driver-truck re-allocations. Differences between the two data collection periods, shown in Table 4, evidence these changes as the usage of less trucks performing trips with trained drivers and instrumented vehicles. Actually, these modifications enhanced the need to remove drivers, vehicles and trips that did not participate in both moments of the analysis, belong to the analyzed routes or have sufficient reliable data to derive in-depth statistical conclusions.

To select the firm's routes with the largest potential for improvement in this study, a Pareto analysis was conducted considering the proportion of trips and the proportion of transported tons. The results are described in Table 5, where the route column presents the origin–destination of the routes using acronyms. The second column provides the average distance in kilometers for all routes connecting the cities, while the third column shows the cargo transported between both cities. The fourth column presents the corresponding proportion for each route regarding total transported cargo. Due to the existent mountainous topography in Colombia, these routes include ascents and descents in the fifth column, where “Δ Height” is an estimate of the altitude difference between origin and destination cities to show the inherent challenge to carry cargo on those trips.

Table 6 presents the profile of drivers who participated in the eco-driving campaign and drove an instrumented vehicle during both periods (baseline and post campaign). Four educational levels were captured (i.e., 1 is the lowest and refers to basic or no education, while 4 is the highest one and refers to undergraduate studies). Table 6 also shows descriptive statistics about driving errors and the on-road data before and after the eco-driving training campaign. The variables related to driving behavior are average speed [km/h] and max speed [km/h] the remainder variables are embedded on driving errors. Only data from trained drivers present at both moments (before and after the training campaign), properly calibrated and completely instrumented vehicles, and confident trips on selected routes are considered in this study.

Table 5
Pareto Routes.

Baseline					Post campaign				
Route	Average distance (km)	Cargo (ton)	%	Δ Height (km)	Route	Average distance (km)	Cargo (ton)	%	Δ Height (km)
Mad-Gir	459.8	411.30	8.09	+1.15	Gir-Bog	512.4	403.82	8.09	+1.20
Gir-Mad	441.3	408.74	8.04	−1.15	Sab-Sop	554.6	714.00	14.3	−1.05
Fun-Gir	495.1	225.50	4.43	−1.15	Sop-Med	550.3	539.37	10.8	−1.05
Gir-Bog	536.2	235.11	4.62	+1.20	Bvt-Bog	557.3	450.96	9.04	+2.60
Sop-Med	615.8	328.68	6.46	−1.05	Gir-Mad	434.9	299.82	6.01	+1.15
Sab-Sop	599.7	374.00	7.35	+1.00	Mad-Gir	475.9	190.00	3.81	−1.15
Car-Med	763.9	303.00	5.96	+1.50	Sop-Cal	613.8	363.09	7.28	−1.55
Sop-Est	620.2	256.90	5.05	+0.75	Fun-Gir	526.5	131.00	2.62	−1.15
Sop-Cal	660.8	162.34	3.19	−1.55	Sop-Est	648.0	334.71	6.71	+0.75

Table 6

Descriptive statistics of the data collected.

Sets	Variables		Average	Standard deviation	Median	Min	Max	
Drivers profiles # of drivers: 18	Age		37.1	2	39	25	52	
	Years of experience		16.2	1.7	15.5	7	30	
	Educational level		2.8	0.2	3	1	4	
Baseline Performance n ₁ : 277 trips Total distance: 158,793 km Total transported: 5069 t	On-road	Total weight [t]	30.58	14.88	32.30	8.5	51.5	
		Load [t]	18.30	10.54	17.00	3.0	34.0	
		Distance [km]	573.26	204.29	524.60	37.0	1676.7	
		Total time [h]	16.99	7.15	15.56	1.56	63.93	
		Driving time [h]	14.21	4.72	13.48	1.2	33.6	
		Average speed [km/h]	40.40	4.64	40.90	18.2	50.8	
		Max speed [km/h]	84.54	6.05	84	75.0	104.0	
	Driving errors	Excessive RPM	#	94.43	137.19	7	0	936
			time [s]	0.15	0.50	0.005	0	4
			Max	2369.31	324.20	2300	1700	3400
		Abrupt acceleration	#	0.99	7.34	0	0	98
			time [s]	0.57	1.89	0	0	12
			max	6.27	5.35	5	3	42
		Abrupt braking	#	0.61	5.18	0	0	70
			time [s]	0.61	1.96	0	0	12
			max	10.14	5.65	9	6	56
		Excessive speed	#	13.94	26.45	2	0	220
			time [s]	0.11	0.57	0.003	0	6
			max	84.09	6.03	83	75	104
		Excessive idle time	#	1.35	2.06	0	0	11
			dur (%)	0.82	1.46	0.13	0	13.6
			time [h]	1.24	0.61	1.16	0.13	4.34
			dur (%)	9.16	4.25	8.55	1.08	22.8
		Post-campaign Performance n ₂ : 240 trips Total distance: 133,719 km Total transported: 4965 t	On-road	Total weight [t]	33.68	16.28	38.66	7.72
Load [t]	20.59			11.60	21.51	2.20	34	
Distance [km]	556.76			143.87	524.65	24.80	1200.2	
Total time [h]	16.29			4.48	15.85	1.53	33.64	
Driving time [h]	14.06			3.71	13.63	0.88	30.50	
Average speed [km/h]	39.95			4.48	39.67	28.13	53.06	
Max speed [km/h]	81.88			4.78	81	75	100	
Driving errors	Excessive RPM		#	96.26	159.06	1	0	684
			max	2401.25	291.22	2300	1900	3200
	Abrupt acceleration		#	0.03	0.25	0	0	3
			max	4.85	1.35	5	3	13
	Abrupt braking		#	0.02	0.14	0	0	1
			max	9.17	2.15	9	6	20
	Excessive speed		#	3.80	8.12	0	0	58
			max	81.35	4.58	80	73	100
	Excessive idle time		#	0.95	1.38	0	0	7
			dur (%)	0.21	3.26	0	0	50.54
			time [h]	1.10	0.86	0.95	0.08	4.83
			dur (%)	0.08	0.06	0.07	0.00	0.26

t: tons, n: sample size, #: number of events, max: maximum values, dur: duration of the error, OGB: outside the green band.

3.1. Changes after the training campaign

The immediate effects after eco-driving training can be observed in [Table 7](#). There is an average FC reduction of 6.8% when it is measured in [l/t-100 km] and a 5.5% reduction when it is measured in [l/100 km-transported tons]. The reduction on the second measurement reflects that load effects are significant. To analyze the significance of FC reduction, hypothesis tests are performed to validate the statistical decreases on the average FC, assuming it is normally distributed. The results show significant reductions in both cases, although a higher variability is observed when the logistics perspective based on [l/100 km-transported tons] is used.

Similarly, [Table 8](#) shows a summary of results to analyze the importance of distinct factors with regards to FC.

The results show that average distance and driving time did not change between the samples representing the moments before and after the eco-driving training. This finding is explained by the observation that the vehicles are non-driving dependent variables. At the same time, average speed did not vary between the periods. However, reductions on most driving errors were found with the exception of excessive engine speed [RPM]. Important reductions, to a maximum of 97%, on the number of driving error events are particularly noteworthy. From this finding, it can be derived that driving errors play an important role in fuel consumption savings and that eco-driving training helps promote safer driving and improves logistics performance.

Table 7
Reduction on fuel consumption.

	Baseline		Post-campaign		Change	
	[l/t-100 km]	[l/100 km-tt]	[l/t-100 km]	[l/100 km-tt]	[l/t-100 km]	[l/100 km-tt]
Average	1.716	3.115	1.600	2.944	− 6.8%	− 5.5%
St dev	0.411	1.115	0.497	1.417	20.9%	27.1%
Median	1.694	2.993	1.565	2.592	− 7.6%	− 13.4%
Min	0.130	0.197	0.401	0.603	208.1%	205.9%
Max	2.963	8.172	3.028	8.599	2.2%	5.2%
N	277	277	239	239	− 13.7%	− 13.7%
$H_1: \mu(\text{post}) < \mu(\text{baseline})$						
t.statistic	2.900	1.530		dof	514	514
p_value	0.002	0.006		s_p	0.453	1.264

St dev: Standard deviation, tt: transported tons, p: significance level, dof: degrees of freedom, s_p : pooled standard deviation.

Table 8
Changes in diverse variables after training.

Variable	Hypothesis testing $H_1: \mu(\text{post}) \text{ vs. } \mu(\text{baseline})$	p_value	% change
Average distance	$\neq$	0.284	− 2.1
Driving time	$\neq$	0.691	− 1.1
Average speed	$\neq$	0.264	− 1.1
Maximum speed	$<$	0.000	− 3.1
# of excessive speed events	$<$	0.000	− 72.7
# of excessive RPM events	$<$	0.880	1.9
# of abrupt acceleration events	$<$	0.015	− 97.1
# of abrupt braking events	$<$	0.030	− 96.6
# of excessive idle time events	$<$	0.004	− 84.3
% of time OGB	$<$	0.000	− 11.4

3.2. Multiple linear regression models

Once the impact of the eco-driving campaign on the factors have been understood, regression analyses are suitable to investigate the interactions among them. In this case, fittings models on $\log(FC)$, with FC measured as [l/t-100 km] and [l/100 km-tt] were built for both data sets: baseline and post-campaign. Regressor variables were divided into three large groups: (a) driver profile (i.e., age, years of experience and educational level), (b) operational variables that are driver-dependent (e.g., speed, acceleration and braking) and independent (e.g., routes and vehicle features), and (c) driving errors that are completely driver dependent (i.e., excessive number of events, duration and magnitude of the observed variables).

Transformations on certain variables are explored by using logarithm functions such as $\log(\text{distance})$, $\log(\text{age})$, inverse functions such as % of time outside green band (OGB), and interactions such as utilization factor determined by total weight/capacity, empty weight/capacity, and duration of driving errors as percentage of total driving time. In addition, the altitude difference between origin city and destination city of any route was also included to distinguish positive altitude differences (i.e., *Ascent Route* = 1 if altitude destination, altitude origin is greater than 0).

Based on the pairwise correlation analyses between fuel consumption and other measured variables, and literature findings, Table 9 shows a summary of four selected regression models that consider the three large groups of variables. A set of statistical criteria were met to guarantee the building of proper statistical models. These criteria consist of the following: (1) high adjusted determination coefficients (R^2a); (2) low variance inflation factors (VIF) to avoid misinterpretations of regression coefficients due to multicollinearity, (3) significance of regressor variables less than 0.05, and (4) inclusion of variables that were consistently present in the models with the best criteria 1, 2 and 3. In general, models sharing the same set of regressor variables for both unit options of y are preferred. We can observe that response variables are explained at least in 73% of the cases by these regression models. The interpretation of the results from Table 9 and other experiments is discussed in the next subsection.

3.3. Discussion of the models

Table A1 in the appendix extends the results obtained from regression models presented in Table 9, as well as other components such as coefficients (betas), standard errors (SE), and VIF. Now centering the discussion on the results presented in Table 9, standardized coefficients are highlighted to provide an idea of the relative importance of the variables included in the model and their relationships (i.e., signs) regarding the FC. It is worth noting that the R^2a values obtained are better than expected, particularly considering the huge uncontrollable variability on routes that represent approximately 760 km and 1–3 days.

Table 9
Summary of regression models on $y = \log(FC)$.

Regression			Baseline				Post-campaign				
			N								
			R ² a								
			FC unit	[l/t-100 km]		l/100 km-tt]		[l/t-100 km]		[l/100 km-tt]	
Regression variables			β	sign	β	sign	β	sign	β	sign	
Driver profile			Years of experience				0.254	+	0.192	+	
Operational	Driv. Indep	Total weight	0.763	−	−0.927	−	−1.001	−	−1.130	−	
		Utilization	−0.233	−	−0.422	−	−0.150	−	−0.341	−	
		Distance	−0.135	−	−0.079	−					
	Driv. Dep.	Ascent route	0.185	+	0.165	+	0.073	+	0.054	+	
		Average speed					−0.384	−	−0.303	−	
Driving errors			# of exc. RPM	0.289	+	0.190	+	0.404	+	0.315	+
			Exc. RPM max					−0.386	−	−0.319	−
			Abr. accel. max	−0.191	−	−0.158	−				
			Abr. brak. max	−0.343	−	−0.265	−				
			# of exc. speed					0.148	+	0.127	+
			# exc. idle time	0.128	+	0.088	+				
			% of time OGB	−0.165	−	−0.081	−	−0.127	−	−0.111	−

FC: fuel consumption, β : standardized coefficient, Driv: Driver, Indep: Independent, Dep: Dependent, Exc: excessive, #: number of events, Max: maximum, accel: acceleration, brak: braking, Abr: Abrupt, OGB: Outside green band, tt: tons transported.

From these analyses, nine of 13 variables are present in the regression models for baseline and the post-campaign periods. Five of these 9 significant variables are shared by models in both periods, and they are total weight, utilization, ascent route indicator, excessive number of RPM, and % of the time OGB. These variables maintain their signs to provide a sense of robustness and consistency to the acquired results. Total weight appears to be the most influential variable on FC followed by the driving error that measures the number of excessive RPM. It is interesting to note the negative sign on total weight. This finding suggests that it is better to schedule large and fully loaded vehicles. This suggestion is also supported by the utilization ratio (TW/Capacity) which is an interesting variable that has negative influence in most of the experiments. Thus, the firm under analyses has opportunities on consolidating better the freight and using the capacity of its working vehicles.

Regarding driver profile, none of the related variables influence FC in the baseline. However, after the eco-driving campaign, experience becomes significant in increasing the FC. This result shows that the most experienced drivers presented higher FC after the training than the other drivers. Delhomme et al. (2013) found that the French senior drivers of private vehicles presented less difficulty adopting eco-driving behavior, and De Abreu e Silva et al. (2015) acquired similar conclusions for bus companies. Despite the conclusions of these studies are different, this finding may reflect the reluctance of senior truck drivers to change their driving styles, or conversely, that junior drivers are more sensitive to the training; behaviors that might be improved by the firm with appropriate incentive or penalty programs.

The results from the speed analysis, with mean values of approximately 40 km/h, suggest that increasing the average speed will improve the FC after the training. Although it could be counterintuitive, Walnum and Simonsen (2015) reported similar conclusions for a Norwegian heavy-duty truck transport company. However, a sustained speed increase is unfeasible due to the current road and topographic conditions of the region.

When considering the slope of the mountains, higher ascent gradients decrease the average speed, which significantly increases the FC. Thus, an average speed of 63 km/h decreased to 30 km/h in the studied routes, showing how the topography highly influences fuel consumption and related effects (e.g., emissions) as also shown by Walnum and Simonsen (2015). From driver independent operational variables, it was observed that longer trips that present less FC and FC will be greater when the final destination is located at a higher altitude point than the origin of the route. This finding means that longer routes are usually more effective due to better vehicle consolidation (i.e., utilization) and the opportunity to offset shorter distance effects such as congestion, multiple stops and not achieving a steady (cruise) speed.

On driving errors, during the baseline period, the maximum values of abrupt accelerations and the resulting abrupt brakings are influential. However, they presented a contrary sign as expected, in contrast to the number of excessive RPM events. In the post-campaign period, the maximum values of excessive RPM appeared to replace acceleration and braking errors in the baseline case with similar effects. Other errors included with the expected sign were the number of violations of maximum speed and the number of excesses in idle times. None of the error measured as durations were relevant; so they were not included in Table 9. Finally, increasing the percentage of time outside green band (% of time OGB) appeared to be beneficial to FC. Thus, the definition of OGB in this context is widely arguable, and it will be future research since it extends beyond the scope of this research.

4. Conclusions and recommendations

From the wide range of strategies and actions that had been developed concerning fuel consumption and energy efficiency in a vehicle fleet, this study focused on eco-driving. This work considered an eco-driving program that was applied to a company's fleet

that transports goods through a Latin American region with hilly topographies; this is a novel condition in this type of studies. The variables analyzed before and after the training campaign were logistics operational variables, road and vehicle features, driving errors and driver profiles, among others.

Fuel consumption (FC) in liters per ton-100 km and liters per 100 kilometer-transported tons were used. The first metric is used from the vehicle perspective, while the second metric is proposed as a metric that can be used from the logistics perspective. After the eco-driving campaign, an FC reduction of 6.8% was obtained when it is measured in [l/t-100 km] and a 5.5% reduction when it is measured in [l/100 km-transported ton].

After the analysis of the variables that were collected from real data, it was determined that the variables that are significant in the FC are, as related to driver profile, years of experience; as related to operational variables which are driver independent, total weight, utilization, distant and ascent route; as related to operational variables which are driver dependent, average speed; and finally, as related to driver errors, number of excessive speed events, number of excessive RPM events, number of abrupt acceleration events, number of abrupt braking events, number of excessive idle time events and percentage of time outside the green band. Results from the particular case study reveal that driver experience, road conditions (i.e., steepness) and driving errors such as excessive speed, excessive revolutions per minute (RPM) and excessive idle time, must be controlled.

From the regression analysis, nine of 13 variables are present in the regression models for baseline and the post-campaign periods. Five of these 9 significant variables are shared by models in both periods; these are total weight, utilization, ascent route indicator, excessive number of RPM, and percentage of the outside green band time. The total weight appears to be the variable that is most influential on FC followed by the driving error, which measures the number of excessive RPM. All of the driver profile variables influence FC in the baseline; after the eco-driving campaign, experience becomes significant to increasing the FC, suggesting that experienced drivers are less sensitive to eco-driving training.

The results suggest that having higher speeds, longer trips, and downstream routes present better FCs. Related to driving errors, during the baseline period, the maximum values of abrupt accelerations and abrupt brakings were influential. The maximum values of excessive RPM in the post-campaign period appeared to replace acceleration and braking errors in the baseline case with similar effects.

Although certain findings are company-specific, these, along with the data collection and the methodology applied help to gain knowledge about the effects on fuel consumption of eco-driving initiatives in Latin America. This finding is part of one of the main contributions of this study given the scarce body of literature in the region by addressing logistics specificities and geographic challenges for long-distance allocations. This study will also help us provide focus to only the relevant variables from a large data set and redefine the means by which the company assessed driver performance, particularly in terms of driving errors control before and after eco-driving training. Further study is required to understand outside of green band concepts and the effect of long routes passing through large cities in fuel consumption.

Thus, considering the increasing adoption of real-time data logger systems by most of the companies, this methodology may be easily replicated to address the research on additional variables relying on the various circumstances, locations and time. Furthermore, richer data sets with controllable variables can be used as inputs for designing corporate strategies for driver assessment, incentive programs, eco-driving effectiveness, and vehicle allocation.

For freight transportation cases characterized by long routes, steep mountain trips, average low speed, and heavy-duty vehicles, the results found in this study serve as a starting point to make the long-term decisions at a company level. The general recommendations are to (a) use the largest vehicles as fully loaded as possible, (b) work preferably with receptive (i.e., non-reluctant, maybe young) drivers, (c) use eco-driving training programs not only to improve fuel consumption but also to improve safe driving habits, (d) use data logger systems for real-time fleet control, and most importantly, and (e) concentrate control efforts in only a few influential variables on significant performance measures, such as controlling excessive vehicle speed, engine revolutions, and idle time and optimizing the vehicle load factor.

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Appendix A

(See Table A1).

Table A1
Output from multiple regression models.

	Driver profile				Operational (Driver independent)				Oper. (Driver dep.)		Driving errors			
	Regression	Y	Const.	Years Exp.	Total Weight	TW/Cap	Distance	Ascent Route	Aver. Speed	# of exc. RPM	Abrupt accelerat. Max	Abrupt braking Max	# of exc. Idle time	% of time OGB
<i>Base line</i> Good	N	277	log(FC) [/ t-100 km]	B	0.79823			0.05065						
	R ²	0.801	SE		0.03931			0.00771						
	R ² a	0.794	β					0.185						
	max sig.	0.012	VIF					1.07						
					2.08	1.49	1.15			2.46	7.64	7.61	1.18	1.7
Good transported-ton	N	277	log(FC) [/ 100 km-tt]	B	1.41171			0.05737						
	R ²	0.870	SE		0.04041			0.00793						
	R ² a	0.866	β					0.165						
	max sig.	0.010	VIF					1.07						
					2.08	1.49	1.15			2.46	7.64	7.61	1.18	1.7
<i>Post campaign</i>														
Good	Regression	Y	Const.	Years Exp.	Total Weight	TW/Cap	Distance	Ascent Route	Aver. Speed	# of exc. RPM	Exc RPM Max	# of exc. Speed	# of exc. Idle time	% of time OGB
	n	239	log(FC) [/ t-100 km]	B	1.69000			0.02487						
	R ²	0.738	SE		0.09303			0.01165						
	R ² a	0.729	β					0.073						
	max sig.	0.034	VIF					1.03						
Good transported-ton	n	239	log(FC) [/ 100 km-tt]	B	2.41317			0.23457						
	R ²	0.831	SE		0.09465			0.01185						
	R ² a	0.825	β					0.054						
	max sig.	0.049	VIF					1.03						
					1.77	1.24				2.44	1.57	1.24		

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